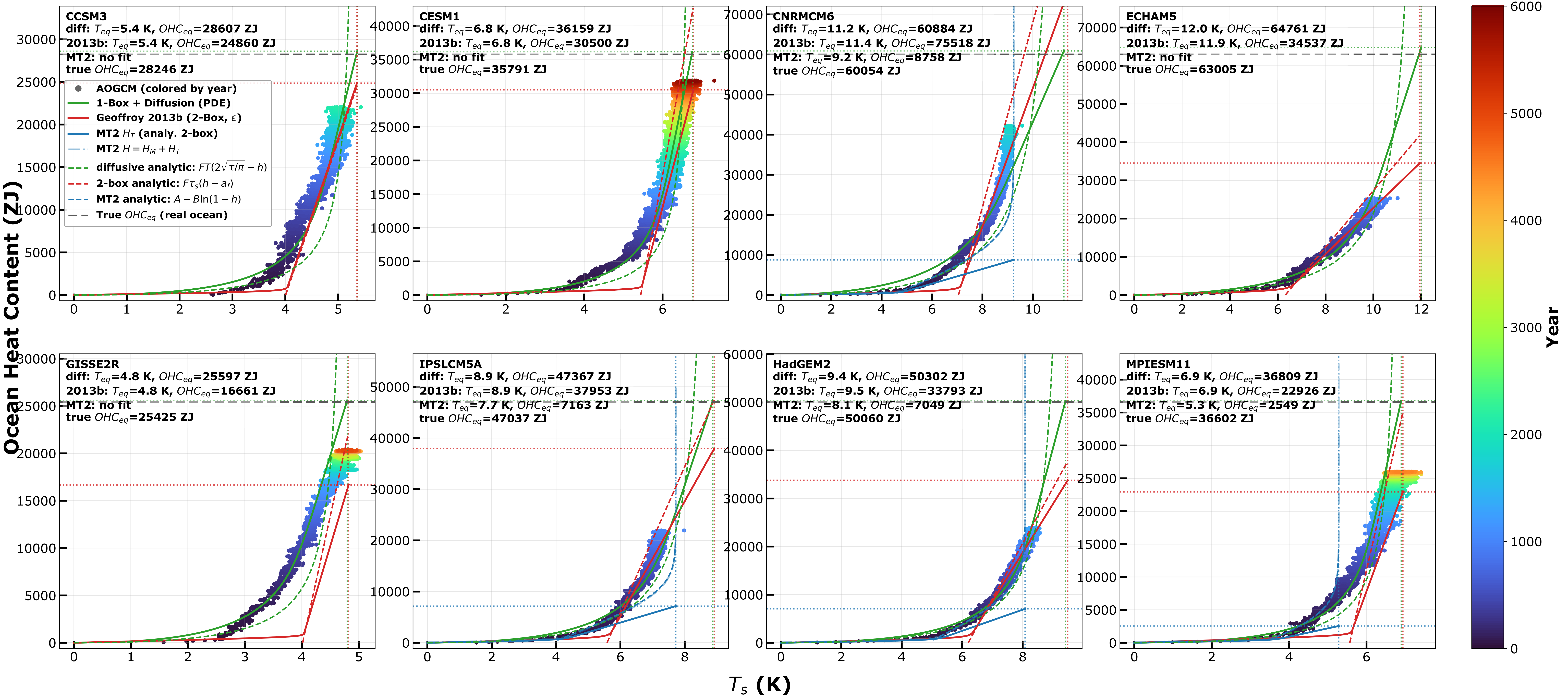
4xCO$_2$ OHC vs. Surface Temp: AOGCMs and three fitted models

03_lrfit-50yravg parameters; MT2 H_T (analy. 2-box) from 01_replicate



Ocean Heat Content (ZJ)
T_s (K)
Year
0
1000
2000
3000
4000
5000
6000
AOGCM (colored by year)
1-Box + Diffusion (PDE)
Geoffroy 2013b (2-Box, ε)
MT2 H_T (analy. 2-box)
MT2 $H = H_M + H_T$
diffusive analytic: $FT(2\sqrt{\tau/\pi} - h)$
2-box analytic: $F\tau_s(h - a_f)$
MT2 analytic: $A - B\ln(1-h)$
True OHC_{eq} (real ocean)
CCSM3
diff: T_{eq}=5.4 K, OHC_{eq}=28607 ZJ
2013b: T_{eq}=5.4 K, OHC_{eq}=24860 ZJ
MT2: no fit
true OHC_{eq}=28246 ZJ
CESM1
diff: T_{eq}=6.8 K, OHC_{eq}=36159 ZJ
2013b: T_{eq}=6.8 K, OHC_{eq}=30500 ZJ
MT2: no fit
true OHC_{eq}=35791 ZJ
CNRMCM6
diff: T_{eq}=11.2 K, OHC_{eq}=60884 ZJ
2013b: T_{eq}=11.4 K, OHC_{eq}=75518 ZJ
MT2: T_{eq}=9.2 K, OHC_{eq}=8758 ZJ
true OHC_{eq}=60054 ZJ
ECHAM5
diff: T_{eq}=12.0 K, OHC_{eq}=64761 ZJ
2013b: T_{eq}=11.9 K, OHC_{eq}=34537 ZJ
MT2: no fit
true OHC_{eq}=63005 ZJ
GISSE2R
diff: T_{eq}=4.8 K, OHC_{eq}=25597 ZJ
2013b: T_{eq}=4.8 K, OHC_{eq}=16661 ZJ
MT2: no fit
true OHC_{eq}=25425 ZJ
IPSLCM5A
diff: T_{eq}=8.9 K, OHC_{eq}=47367 ZJ
2013b: T_{eq}=8.9 K, OHC_{eq}=37953 ZJ
MT2: T_{eq}=7.7 K, OHC_{eq}=7163 ZJ
true OHC_{eq}=47037 ZJ
HadGEM2
diff: T_{eq}=9.4 K, OHC_{eq}=50302 ZJ
2013b: T_{eq}=9.5 K, OHC_{eq}=33793 ZJ
MT2: T_{eq}=8.1 K, OHC_{eq}=7049 ZJ
true OHC_{eq}=50060 ZJ
MPIESM11
diff: T_{eq}=6.9 K, OHC_{eq}=36809 ZJ
2013b: T_{eq}=6.9 K, OHC_{eq}=22926 ZJ
MT2: T_{eq}=5.3 K, OHC_{eq}=2549 ZJ
true OHC_{eq}=36602 ZJ